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Clinical Practice Guideline:
Emergency evaluation and management of patients with known
familial hypokalemic periodic paralysis (HypoKPP)

I. Introduction

The purpose of this guideline is to facilitate and establish a framework for any child who presents to the Pediatric Emergency Room at Kentucky Children's Hospital with a known diagnosis of *familial hypokalemic periodic paralysis*.

Hypokalemic periodic paralysis is a rare, autosomal dominant channelopathy leading to muscle weakness and paralysis with low serum potassium levels related to a genetic defect in a voltage gaited calcium channel. Affected individuals shift potassium intracellularly during attacks leading to cough and profound hypokalemia. Patients with this disease are **not** body potassium depleted, which makes the management approach more complicated as they are at risk for rebound hyperkalemia.

II. Inclusion criteria

This CPG can be used for any patient with a known diagnosis of familial hypokalemic periodic paralysis who presents to the Emergency Room with hypokalemia. As for any clinical practice guideline, this can certainly only be of frame work and the therapy needs to be tailored to the individual clinical situation.

III. Diagnostic Evaluation

1. Family history of hypokalemic periodic paralysis should be confirmed and a careful examination to assess degree of muscle weakness and paralysis should be conducted
2. Patient should be placed on telemetry
3. Pediatric nephrologist on-call should be notified to be involved in the care of the patient
4. All patients who present to the emergency room at Kentucky Children's Hospital should have a STAT renal function panel and magnesium level drawn.
5. Other tests might be indicated based on clinical circumstances. This might include hemogram, thyroid functions, urinalysis, EKG, or others.

IV. <u>Differential diagnosis</u>

Other hypokalemic conditions might present with paralysis or muscle weakness and it is therefore important to obtain a complete family history to confirm the diagnosis.

V. <u>Treatment</u>

As patients with hypokalemic periodic paralysis are usually rather easily identified given the family history and previous episodes (most patients are known to the ED physicians), rapid initiation of treatment is of most importance. All interventions are initiated with the intention to improve the muscle weakness and paralysis of the patient and to prevent severe hypokalemia.

1. A peripheral IV should be placed immediately and blood should be drawn for the test outlined above.
2. Pending the initial serum potassium level, an infusion of 10 mEq of potassium chloride IV (in non-Dextrose containing solution) over 30 minutes as well as oral administration of 40 mEq of potassium chloride should be initiated.
3. A second dose of IV 10 mEq of potassium chloride should be ordered to the bedside with the initial orders.
4. After the first infusion of potassium chloride and the first dose of oral potassium is complete, the following situations might be encountered:
    a. The serum potassium level has not been reported yet: Initiate second IV infusion of 10 mEq of potassium chloride and another 40 mEq of potassium chloride by mouth until the serum potassium level is back if the patient is still symptomatic. Should the patient reported improvement or even resolution of symptoms, wait for serum potassium level to be reported.
    b. The serum potassium level is back and is less than 3 mmol/L:

Repeat 10 mEq of IV potassium chloride as well as 40 mEq of oral potassium chloride. Draw another serum potassium level STAT. Reevaluate patient with a thorough physical examination to see if the muscle weakness and paralysis are improving. Continue procedure until serum potassium level is greater than 3 mmol/L. Once the level is greater than 3 mmol/L, continue with CPG under 4 c.

c. The serum potassium level comes back and is greater than 3 mmol/L:
   Give another 40 mEq of oral potassium chloride but no further IV potassium chloride. Reevaluate patient with a thorough physical examination to see if the muscle weakness and paralysis are improving. Keep patient for observation over the next 2-3 hours and repeat a potassium level in 2 hours. If the level is within the reference range (3-5.5 mmol/L) and the patient is improving symptomatically, patient can be discharged from the emergency room.
d. Depending on the clinical course and the amount of potassium that was given as well as the last serum potassium level, hospital admission for observation might need to be considered. This will be discussed an individualized basis between the pediatric emergency medicine attending and pediatric nephrologist.

VI. Follow-up

Patients with hypoKPP are routinely followed through the pediatric nephrology clinic at Kentucky Children's Hospital. We routinely advise the family to call the pediatric nephrologist on call at 859-218-1675 during the day and at 859-323-5321 doing after hours to let him/her know that the patient is on the way to the pediatric emergency room. In addition, patients with hypoKPP are seen in regular intervals in the pediatric nephrology outpatient clinic.

VII. References

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